Chain Rules [ใช้กับโจทย์ที่มองเป็นก้อนได้]

สูตร — (diff นอก)(diff ไส้)

① $y = (2x+1)^2$

method $2(2x+1)^2 \cdot 2$

$= (4x+2) \cdot 2$

$= 8x + 4$ #

② $\sqrt{x^3 + 7x + 1}$

method $(x^3 - 7x + 1)^{\frac{1}{2}}$

$= \frac{1}{2}(x^3 - 7x + 1)^{-\frac{1}{2}} \cdot (3x - 7)$

$= \frac{1}{2} \cdot \frac{3x^2 - 7}{\sqrt{x^3 - 7x + 1}}$ #

Exponential [diff ยกกำลัง]

สูตร — $e^x = e^x$

$e^u = e^u(u')$

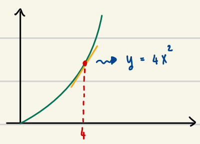
① $\frac{d}{dx}[e^{\frac{-3}{x}}]$

method $\left[\frac{3}{x^2}\right] e^{\frac{-3}{x}} = \frac{3e^{\frac{-3}{x}}}{x^2}$

Note! $\frac{-3}{x^1} \frac{d}{dx} = 3x^{-1} = 3x^{-2} = \frac{3}{x^2}$

Slope

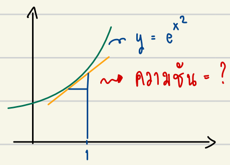
$m = \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$

① 

$m = \frac{dy}{dx} = \frac{d}{dx}(4x^2)$

$m = 8x$

$\therefore m = 8(4) = 32$

② 

$m = \frac{dy}{dx} = \frac{d}{dx}(e^{x^2})$ ตัว u

$= e^{x^2} \cdot 2x$

$m = e^{1^2} \cdot (2(1)) = 2e \;/\; \approx 5.437$ #

Definite Integral [อินทิเกรตจำกัดเขต]

$\int_a^b f(x)\,dx$ — การกำหนดขอบเขต $a \rightarrow b$

การหาค่า :

$\int f(x)\,dx = F(x) + c$

$\therefore \int_a^b f(x)\,dx = F(b) - F(a)$ → ได้พื้นที่ใต้กราฟ

Note! วิธีคือ 1. Integrate 2. แทนค่า 3. ลบกัน

① $\int_{-1}^{2} 3x^2\,dx$

method $\int 3x^2\,dx = \frac{\cancel{3}x^3}{\cancel{3}} + c$

Note! How to Integrate : $\frac{x^{n+1}}{n+1} + c$ constant ค่าคงที่

$\int_{-1}^{2} 3x^2\,dx = [x^3]_{-1}^{2}$

① — แทน $x = 2$: 2^3

② — แทน $x = -1$: $(-1)^3$

① − ② $= 2^3 - (-1)^3$

$= 8 + 1 = 9$ #

② $\int_1^8 \sqrt[3]{x}\,dx$

method $\int x^{1/3}\,dx = \frac{x^{4/3}}{\frac{4}{3}} + c = \frac{3}{4}x^{4/3} + c$

$\int_1^8 \sqrt[3]{x}\,dx = [\frac{3}{4}x^{\frac{4}{3}}]_1^8$

① — แทน $x = 8$: $\frac{3}{4}(8)^{\frac{4}{3}}$

② — แทน $x = 1$: $\frac{3}{4}(1)^{\frac{4}{3}}$

① − ② $= \frac{3}{4}(8)^{\frac{4}{3}} - \frac{3}{4}(1)^{\frac{4}{3}}$ $(2^{\cancel{3}})^{\frac{4}{\cancel{3}}} = 2^4$

$= \frac{3}{\cancel{4}}(\cancel{16}^{\,4}) - 1$

$= 12 - 1 = 11$ #

③ $\int_1^4 \sqrt{x}\,(1+2\sqrt{x})\,dx$

method $\int \sqrt{x}\,(1+2\sqrt{x})\,dx = \int \sqrt{x}+2x\ dx$

$= \int x^{\frac12}+2x\ dx$

$= \frac{x^{\frac32}}{\frac32} + \frac{\cancel{2}x^2}{\cancel{2}} + C = \frac23 x^{\frac32} + x^2 + C$

$\int_1^4 \sqrt{x}\,(1+2\sqrt{x})\,dx = \left[\frac23 x^{\frac32} + x^2\right]_1^4$

① — แทน x = 4 : $\frac23\cdot 4^{\frac32} + 4^2$

② — แทน x = 1 : $\frac23 + 1$

① - ② $= \left(\frac23\cdot 4^{\frac32} + 4^2\right) - \left(\frac23 + 1\right)$

$= \frac{64}{3} - \frac53$

$= \frac{59}{3}$ #

L'hospital Rule [การหาลิมิตด้วยโลปิตาล]

$\lim_{x\to\Delta}\frac{f(x)}{g(x)}$ => interminate form

Note! interminate form : $\frac00,\ \frac{\infty}{\infty},\ 0\times\infty,\ 1^{\infty}$

$\lim_{x\to\Delta}\frac{f(x)}{g(x)} = \lim_{x\to\Delta}\frac{f'(x)}{g'(x)}$ => diffบน / diff ล่าง

① $\lim_{x\to 1}\frac{x^2-1}{x-1} = \frac{1^2-1}{1-1} = \frac00$

method $\frac{2x}{1} = 2(1) = 2$ #

② $\lim_{t\to\infty}\frac{\ln(3t)}{t^2} = \frac{\ln(3(\infty))}{\infty^2} = \frac{\infty}{\infty}$

method $\frac{1}{\cancel{3}t}\cdot\cancel{3} = \frac{1/t}{2t}$

$\lim_{t\to\infty} = \frac{1}{2t^2} = \frac{1}{\infty} = 0$ #

③ $\lim_{x\to -4}\frac{\sin(\pi x)}{x^2-16} = \frac{\sin(-4\pi)}{16-16} = \frac00$

method $\frac{\cos(\pi x)\cdot\pi}{2x}$

$\lim_{x\to -4}\frac{\cos(\pi x)\cdot\pi}{2x}$

$= \frac{\pi\ \cancel{\cos(-4\pi)}^{\,1}}{2(-4)}$

$= \frac{\pi}{-8}$ #

Substitution technic [เทคนิคแทนตัวแปร]

Note! ทำให้ Integrate ง่ายขึ้น

① $\int \sin(\overset{u}{3x+1})\,dx$

method $u = 3x+1 \rightarrow \frac{du}{dx} = 3 \rightarrow dx = \frac{du}{3}$

$\int \sin u\,\frac{du}{3} = \frac13\int \sin u\,du = -\frac13\cos u + C = -\frac13\cos 3x+1 + C$ #

② $\int x(2x^2+3)^{10}\,dx$

method $u = 2x^2+3 \rightarrow \frac{du}{dx} = 4x \rightarrow dx = \frac{du}{4x}$

$\int \cancel{x}\,u^{10}\,\frac{du}{4\cancel{x}} = \frac14\int u^{10}\,du = \frac14\,\frac{u^{11}}{11} + C = \frac14\,\frac{(2x^2+3)^{11}}{11} + c$

③ $\int 2x^2\sqrt{x^3-4}\,dx$

method $u = x^3-4 = \frac{du}{dx} = 3x^2 = dx = \frac{du}{3x^2}$

$\int \cancel{2x^2}(u)^{\frac{1}{2}}\frac{du}{3\cancel{x^2}} \to \frac{2}{3}\int u^{\frac{1}{2}}du = \frac{2}{3}\frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{4}{9}u^{\frac{3}{2}} + c = \frac{4}{9}(x^3-4)^{\frac{3}{2}} + c$ #

Integrate Exponential

$\int a^x dx = \frac{a^x}{\ln a} + c$

Note! $\int e^x dx = \frac{e^x}{\ln e} + c = e^x + c$ #

$\ln e = 1$

① $\int_1^3 2^x dx$

method $\int_1^3 2^x dx = \left[\frac{2^x}{\ln 2}\right]_1^3$

① — แทนค่า $x = 3$: $\frac{2^3}{\ln 2}$

② — แทนค่า $x = 1$: $\frac{2^1}{\ln 2}$

① − ② = $\frac{2^3}{\ln 2} - \frac{2}{\ln 2} = \frac{6}{\ln 2}$ #

② $\int x\cdot 3^{x^2+1}dx$

method $u = x^2+1 \to \frac{du}{dx} = 2x \to dx = \frac{du}{2x}$

$\int \cancel{x}\cdot 3^u\frac{du}{2\cancel{x}} = \frac{1}{2}\int 3^u du = \frac{1}{2}\frac{3^u}{\ln 3} = \frac{1}{2}\frac{3^{x^2+1}}{\ln 3} + c$ #

③ $\int x^2 e^{x^3+1}dx$

method $u = x^3+1 \to \frac{du}{dx} = 3x^2 \to dx = \frac{du}{3x^2}$

$\int \cancel{x^2}e^u\frac{du}{3\cancel{x^2}} = \frac{1}{3}\int e^u du = \frac{1}{3}\frac{e^u}{\ln e} = \frac{1}{3}e^{x^3+1} + c$ #

diff tregonometry

sin / cos / tan | cosec / sec / cot

$\frac{1}{\sin}$, $\frac{1}{\cos}$, $\frac{1}{\tan}$

$\frac{d}{dx}\sin x = \cos x$ | $\frac{d}{dx}\cos x = -\sin x$

① $\frac{d}{dx}\sin(x^2-2x)$

method $\cos(x^2-2x)\cdot(2x-2)$ # # Chain Rule!

② $\frac{d}{dx}(\sin(x^2-2x+1)\cdot\cos^2x)$

$u = \cos^2 x \to -2\cdot\cos x\cdot\sin x$ # Chain Rule

method $[\sin(x^2-2x+1)\cdot(-2\cdot\cos x\cdot\sin x)] + [\cos^2x\cdot(\cos(x^2-2x+1)\cdot(2x-2))]$ # Product Rule! [หน้า·หลัง'] + [หลัง·หน้า']

indeterminate form

$[\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, 1^\infty]$

1) แยกตัวประกอบ

2) Conjugate

3. L'hospital

1) แยกตัวประกอบ

① $\lim_{x\to 3} \frac{x^2+x-12}{3-x} = \frac{3^2+3-12}{3-3} = \frac{0}{0}$

method $\frac{(x+4)\cancel{(x-3)}}{\boxed{3-x}} \leftarrow -\cancel{(x-3)} = \lim_{x\to 3} -x+4 = -3+4 = -7$ #

2) Conjugate # ตัวคูณที่ทำให้รูทหาย

① $\lim_{x\to 0} \frac{x}{2-\sqrt{x+4}} = \frac{0}{2-2} = \frac{0}{0}$

method $\frac{x}{2-\sqrt{x+4}} \cdot \frac{2+\sqrt{x+4}}{2+\sqrt{x+4}} = \lim_{x\to 0} \frac{x(2+\sqrt{x+4})}{2^2-(\sqrt{x+4})^2} = \lim_{x\to 0} \frac{\cancel{x}(2+\sqrt{x+4})}{4-(x+4)} \to \cancel{4}-\cancel{x}-\cancel{4} = \lim_{x\to 0} -(2+\sqrt{x+4}) = -2-\sqrt{0+4} = -2-2 = -4$ #

find limit with วิธีแทนค่า

① $\lim_{x\to 0} \frac{\sin x}{x} = \frac{\sin 0}{0} = \frac{0}{0}$ # สมมติ x ขึ้นมาด้วยค่าที่เข้าใกล้ 0 มากๆ * need calculator * !

method สมมติ $x = 10^{-9}$ $= \frac{\sin 10^{-9}}{10^{-9}} = 1$ #

② $\lim_{x\to 3} \frac{1-\cos(x-3)}{x^2-x-6} = \frac{0}{0}$

method แทน $x \to 3$ มากๆ $= 3+10^{-9} = 0$ # # use calculator ONLY !

Partial derivative [อนุพันธ์ย่อย]

↳ derivative ของ function หลายตัวแปร

$y = f(x_1, x_2, x_3)$

$\frac{\partial y}{\partial x_1}, \frac{\partial y}{\partial x_2}, \frac{\partial y}{\partial x_2}$ => ก.เปลี่ยนแปลงของ y เทียบกับ x แต่ละตัว = partial derivative

Note ! ถ้าเทียบก.เปลี่ยนแปลงของ y กับ $x_1 \to x_2, x_3$ ไม่เปลี่ยน (เป็นค่าคงที่)

① $f = x_1^2 + 2x_2x_1 - 4x_2^2$

method $\frac{\partial f}{\partial x_1} = \frac{\partial}{\partial x_1}(x_1^2 + 2x_2x_1 - 4x_2^2)$

$= 2x_1 + 2x_2(1) - 0$

$= 2x_1 + 2x_2$ #

② $f(x,y) = 3x^5y^{10} + 2y^3 - x^4$ find $f_x(x,y)$, $f_y(x,y)$

method $\frac{\partial}{\partial x}[3x^5y^{10} + 2y^3 - x^4] = 15x^4y^{10} + 0 - 4x^3$ #

$\frac{\partial}{\partial y}[3x^5y^{10} + 2y^3 - x^4] = 3x^5 10y^9 + 6y^2 - 0$ #

Extrema

1) หา Critical Number
→ $f'(x)$
→ เอา $f'(x)$ มา $= 0$
2) แทน Critical Number ใน $f(x)$
3) แทนค่าซ้าย-ขวาใน $f(x)$
4) เปรียบเทียบค่าที่แทนทั้งหมดเพื่อหาค่า MAX , MIN

Note ! [m , n] → ช่วงปิด, Continue = หา MAX , MIN ได้
(m , n) → ช่วงเปิด, Continue = หา MAX ไม่ได้
[m , n] → ช่วงปิด, ไม่ Continue = หา MIN ไม่ได้

① $f(x) = 3x^4 - 4x^3$ on the interval $[-1, 2]$

method $f'(x) = 12x^3 - 12x^2$

$f'(x) = 12x^3 - 12x^2 = 0$ # เอามา = 0

$= 12x^2(x-1) = 0$ # แยก factor

$x = 0, 1$ # Critical Number

$$\left.\begin{array}{llll} f(-1) & \text{แทน Left} & = 7 & \\ f(2) & \text{แทน Right} & = 16 & \text{\# MAX} \\ f(0) & \text{c-number} & = 0 & \\ f(1) & \text{c-number} & = -1 & \text{\# MIN} \end{array}\right\} \text{เทียบเพื่อหา MIN / MAX}$$

Continuity

① $f(x) = \begin{cases} 2x+1 & , x < 2 \\ 5 & , x = 2 \\ 3x-1 & , x > 2 \end{cases}$ $f(x)$ Continue on $(0,4)$?

method $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} [2x+1]$: แทน $x = 2$ $= 2(2)+1 = 5$

$\lim_{x \to 2^+} [3x-1]$: แทน $x = 2$ $= 3(2)-1 = 5$

$f(x)$ เมื่อ $x = 2$ $= 5$

$\therefore$ ทุกจุดเท่ากัน = Continuity #

Note! $f(x)$ continue on $[a,b]$
1) continue on (a,b) ?
2) $f(a) = \lim_{x \to a^+} f(x)$

$f(x)$ continue on $(a,b]$
1) continue on (a,b) ?
2) $f(b) = \lim_{x \to b^-} f(x)$

② $f(x) = \begin{cases} x^3 - 4x & ; x \leq -1 \\ 2x + 5 & ; x > -1 \end{cases}$ Continue $[-1, 3)$?

method 1) $(-1, 3)$ Continue
2) $f(-1) = \lim_{x \to -1^+} f(x)$

$\therefore f(-1) = (-1)^3 - 4(-1) = -1 + 4 = 3$

$\lim_{x \to -1^+} (2x+5) = 2(-1) + 5 = 3$

$\therefore$ ทุกจุดเท่ากัน = Continue on $[-1, 3)$

Implicit diff

การหา $\frac{dy}{dx}$ เมื่อ $f(x,y) = 0$

① $x^2 + y^2 = 2$ $\frac{dy}{dx} = ?$

method $\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(2)$

$2x + 2y \cdot \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{2x}{2y} = \frac{-x}{y}$ #

② $2y^3 + 4x^2 - y = x^6$ หา $\frac{dy}{dx}$

<u>method</u> $\frac{d}{dx}(2y^3 + 4x^2 - y) = \frac{d}{dx}(x^6)$

$$6y^2\frac{dy}{dx} + 8x - \frac{dy}{dx} = 6x^5$$

$$\frac{dy}{dx}[6y^2 - 1] = 6x^5 - 8x$$

$$\frac{dy}{dx} = \frac{6x^5 - 8x}{6y^2 - 1} \quad \#$$